

Onboarding

Julia on HLRS Cluster/Laptops

HLRS Laptops

Most of the course can be completed on the laptops.

Equipped with NVIDIA GPU

Jupyter + VS Code

jupyter lab
code

Course materials

[\\$HOME/JuliaHLRS26](#)

Julia VS Code integration via extension.

The screenshot displays the VS Code interface with the Julia extension installed. The Explorer sidebar on the left shows the project structure, including folders like `.github`, `.vscode`, `benchmark`, `bin`, `deps`, `docs`, `src`, `test`, `.gitignore`, `.travis.yml`, `appveyor.yml`, and `LICENSE.md`. The Julia Explorer sidebar shows the workspace structure, including folders for `Core`, `InteractiveUtils`, `PlotThemes`, `Plots`, and a `Julia REPL`. The main editor shows a Julia script with the following code:

```
9      ...end
10      ...return maxiter
11  end mandel (generic function with 1 method)
12
13  for i in 1:10
14      ...println(i)
15  end
16
17  map
```

A tooltip for the `map` function is visible, showing its signature `map(f, c...) -> collection` and a description: "Transform collection `c` by applying `f` to each element. For multiple collection arguments, apply `f` elementwise." It also provides examples:

```
julia> map(x -> x * 2, [1, 2, 3])
3-element Array{Int64,1}:
 2
```

The bottom panel shows the Julia REPL with the output of the script:

```
4
5
6
7
8
9
10
```

The status bar at the bottom indicates the current file is `sp/isdeflocals*`, the environment is `Julia env: v1`, and the cursor is at `Ln 17, Col 4` with `Spaces: 4` and `UTF-8 LF` encoding.

Let's get started!

- `cd JuliaHLRS26`
- `git pull`
- `jupyter lab`
- `exercises/Day1/1_onboarding/laptop.md`

HLRS Training Cluster

The cluster has two types of nodes.

CPU nodes

“skl”

2 CPUs (skylake)

40 cores total

GPU nodes

“clx-ai”

2 CPUs (cascade lake)

36 cores total

8x NVIDIA V100

Jobs are scheduled with PBS Pro.

Submit a job:

```
qsub job_script.sh
```

Check on your queued/running jobs:

```
qstat -nw
```

VS Code → HLRS Cluster

VS Code → Cluster via SSH

Login node

Works fine, just connect to

`accountname@training.hlr.de`

Compute node

At HLRS, possible but inconvenient

`SetEnv PBS_JOBID=...`

`SSH ProxyJump`

To get Julia, load the necessary system modules.

Modules on the HLRS training cluster

```
module use juliahpc  
module use openmpi  
# shortcut: ml juliahpc openmpi
```

Outside of the course: If there is no (working) system module, use standard binaries provided by [juliaup](#).

Comment: Julia depot is on the parallel file system.

`JULIA_DEPOT_PATH` = where Julia stores stuff
packages
binary dependencies
...

Why not `$HOME`?

Quotas

Can be read-only for compute jobs

Comment: Julia VSCode extension requires a wrapper.

```
[...]
```

```
# Load modules
```

```
module load juliahpc
```

```
module load openmpi
```

```
# Pass on all arguments
```

```
exec julia "${@}"
```

Let's try it!

→ [exercises/Day1/1_onboarding/cluster.md](#)

